WIRESHARK SOC LAB

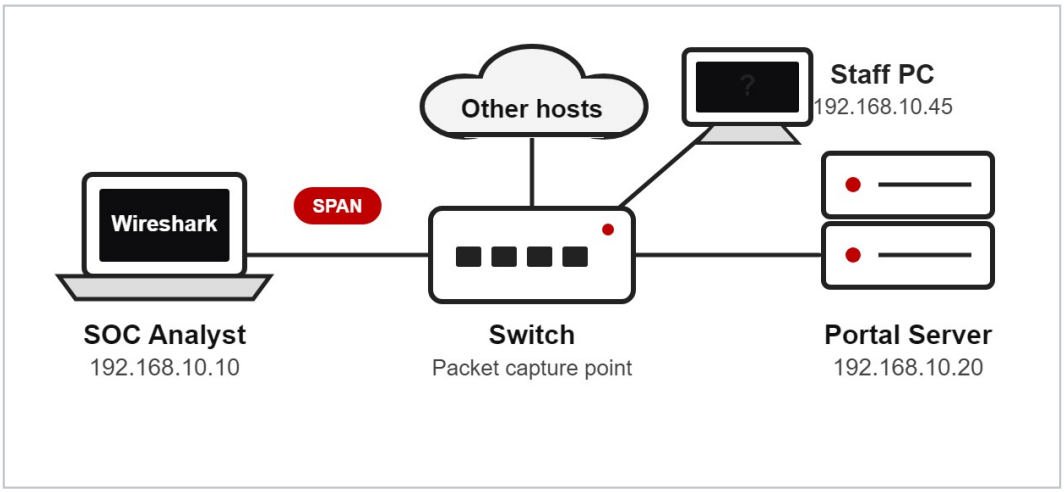
Lab 6 HTTP plain text credentials

Online lab: https://networkwalks.com/wireshark-lab6/

SCENARIO

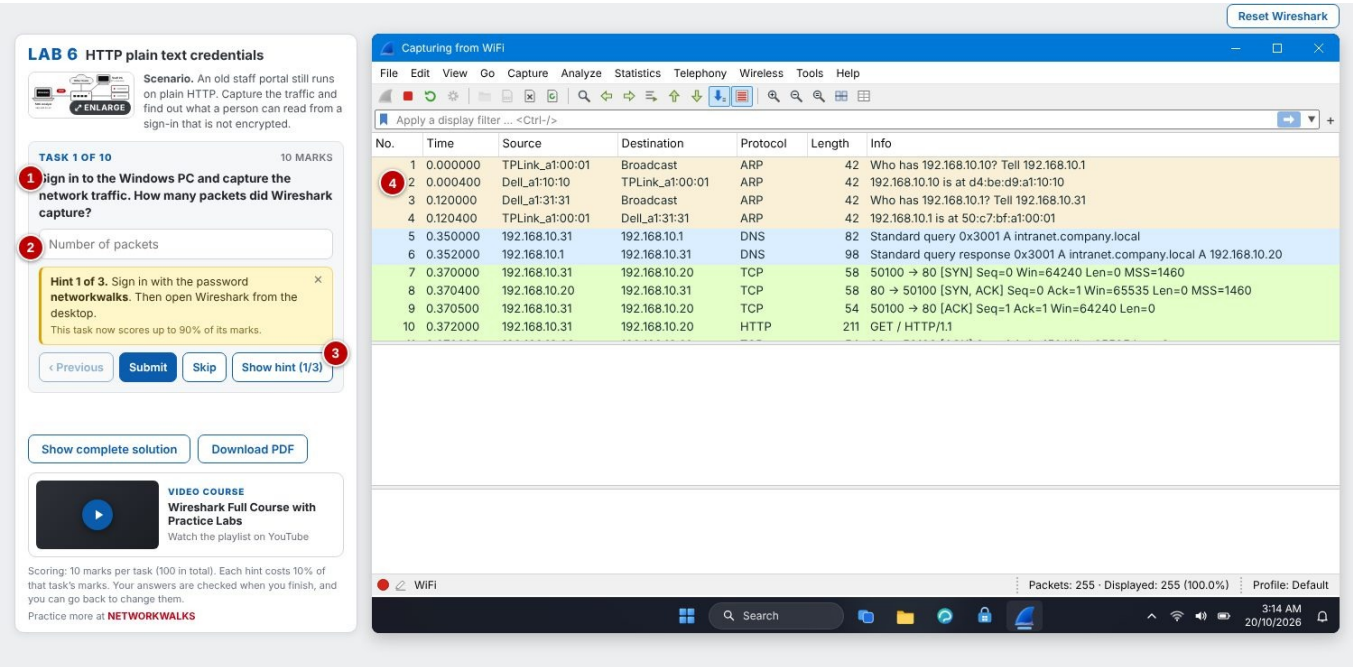
You are a SOC analyst. An old staff portal still runs on plain HTTP. Capture the traffic and find out what a person can read from a sign-in that is not encrypted.

TOPOLOGY DIAGRAM



LAB INSTRUCTIONS

How to complete this lab



The lab screen: the task box on the left, the Windows 11 PC with Wireshark on the right.

1. Read the scenario and the task in the left box (1). Do one task at a time.
2. On the right PC, sign in with the password hint. Double-click Wireshark, then double-click the WiFi interface to start the capture (4).
3. Use the filter bar, the packet list and the Statistics menu to find the answer for the task.
4. Type or choose your answer in the box (2), then press Submit. Use Skip to leave a task for later and Previous to go back. Answers are checked at the end.
5. Stuck? Press Show hint (3). Each hint costs 10% of the marks of that task, only the first time you open it. A task with no answer scores 0.
6. On task 10 press Submit, add a comment if you like, then press Finish. You get a result card that you can download as a picture and share. Show complete solution is always available.

TASKS

Sign in to the Windows PC (user Student, the password hint is on the sign-in screen), start a capture in Wireshark and complete the 10 tasks below. Each task is worth 10 marks (100 in total). Each hint costs 10% of that task's marks.

Task 1. Capture the traffic — 10 marks

Sign in to the Windows PC and capture the network traffic. How many packets did Wireshark capture?

Hint 1: Sign in with the password networkwalks. Then open Wireshark from the desktop.
Hint 2: Double-click the interface that shows an activity graph.
Hint 3: Read the number after Packets: in the status bar at the bottom of the Wireshark window.

Answer: 255

Solution: Sign in with the password networkwalks, double-click Wireshark on the desktop, then double-click the WiFi interface. The status bar shows Packets: 255.

Task 2. Which protocol — 10 marks

A staff member signed in to the old staff portal. Which protocol carried the sign-in?

Options: HTTP | HTTPS | FTP | SSH

Hint 1: Open Statistics › Protocol Hierarchy and look at the application protocols.
Hint 2: Or filter http and look for a packet that mentions /login.

Answer: HTTP

Solution: Filter

```
http
```

. The sign-in packets say POST /login HTTP/1.1. The protocol is HTTP, which is not encrypted.

Task 3. The method — 10 marks

A web form sends the typed values to the server. Which HTTP method did the sign-in use?

Options: GET | POST | PUT | DELETE

Hint 1: Type http.request.method == "POST" in the filter bar and press Enter.
Hint 2: Read the first word in the Info column of the packets for /login.

Answer: POST

Solution: Filter

```
http.request.method == "POST"
```

. The Info column shows POST /login. The method is POST.

Task 4. The login page

10 marks

Which page path did the form send the sign-in to?

Options: /index.html | /login | /admin | /portal

Hint 1: Type http.request.method == "POST" in the filter bar and press Enter.
Hint 2: Read the Info column. The path comes after the word POST.

Answer: /login

Solution: Filter

```
http.request.method == "POST"
```

. The Info column shows POST /login. The path is /login.

Task 5. The PC that signed in

10 marks

Which PC sent the sign-in?

Options: 192.168.10.10 | 192.168.10.31 | 192.168.10.45 | 192.168.10.52

Hint 1: Type http.request.method == "POST" in the filter bar and press Enter.
Hint 2: Read the Source column of the POST packets.

Answer: 192.168.10.45

Solution: Filter

```
http.request.method == "POST"
```

. The Source column shows 192.168.10.45.

Task 6. How many attempts

10 marks

How many times did the user send the sign-in form?

Options: 1 | 2 | 3 | 4

Hint 1: Type http.request.method == "POST" in the filter bar and press Enter.
Hint 2: Count the packets that stay in the list.

Answer: 2

Solution: Filter

```
http.request.method == "POST"
```

. Two packets stay in the list. The user tried 2 times.

Task 7. The username

10 marks

Click one POST packet. In the packet details, open HTML Form URL Encoded. What is the username?

Options: admin | farhan.k | nadia.s | guest

Hint 1: Type http.request.method == "POST" in the filter bar and press Enter.
Hint 2: Click a POST packet. In the middle pane, click the arrow next to HTML Form URL Encoded.

Answer: farhan.k

Solution: Click a POST packet and open HTML Form URL Encoded. The line

```
username = farhan.k
```

shows the user.

Task 8. The password

10 marks

The second sign-in worked (the page said “Welcome back”). What password did the user type that time?

Hint 1: Type http.request.method == "POST" in the filter bar and press Enter.
Hint 2: Click the second POST packet and open HTML Form URL Encoded. You can also right-click it and Follow › TCP Stream.

Answer: Mango#2026

Solution: Click the second POST packet and open HTML Form URL Encoded. It shows

```
password = Mango#2026
```

. The first try (Mango2025) failed.

Task 9. Packet details line

10 marks

Which line in the packet details shows the typed username and password in an easy list?

Options: HTML Form URL Encoded | Line-based text data | Transmission Control Protocol | Frame

Hint 1: Type http.request.method == "POST" in the filter bar and press Enter.
Hint 2: Click a POST packet and scroll to the bottom of the middle pane.

Answer: HTML Form URL Encoded

Solution: At the bottom of the packet details you find HTML Form URL Encoded. It lists the username and password.

Task 10. The fix

10 marks

How can the company stop people from reading passwords like this?

Options: Use HTTPS | Use a longer password | Hide the login page | Use port 8080

Hint 1: Look at the safe sign-in from 192.168.10.31. It used port 443 and you cannot read it.
Hint 2: The safe traffic is encrypted.

Answer: Use HTTPS

Solution: HTTPS encrypts the traffic, so a sniffer sees only unreadable data. The answer is Use HTTPS.

SUMMARY

Protocol:	HTTP, POST /login
User and PC:	farhan.k from 192.168.10.45
Logins:	2 (1 failed, 1 worked)
Password:	Mango#2026
Fix:	Use HTTPS

Watch the Wireshark video playlist: https://www.youtube.com/playlist?list=PLG-wUGWU39ro